# SRM Institute of Science and Technology

Course Code: 21CSC302J | Course Name: COMPUTER NETWORKS

## Assignment 1

### *Unit-1: Introduction to Networks*

**Instructions:** Answer all questions. Draw neat diagrams wherever necessary. Show complete working for numerical questions.

**Q1.** Define a computer network and explain its key objectives and components.

**Q2.** Differentiate between LAN, MAN, PAN and WAN with real-life examples of each.

**Q3.** Compare BUS, STAR, RING, MESH and HYBRID network topologies. Draw a neat diagram for each and list one advantage and one disadvantage.

**Q4.** Differentiate between Circuit Switching and Packet Switching, with suitable examples of each.

**Q5.** Explain the OSI Layered Architecture. Describe the function of each of the seven layers with one example protocol/device per layer.

**Q6.** Compare the OSI reference model with the TCP/IP model, highlighting at least three similarities and three differences.

**Q7.** Explain the functions of the Physical layer. Define the terms Latency, Bandwidth and Delay, and state how they affect network performance.

**Q8.** Compare Twisted Pair cable, Coaxial cable and Fiber Optic cable (guided media) in terms of cost, bandwidth, attenuation and typical usage.

**Q9.** Explain Unguided media - Radio waves, Microwaves and Infrared - with their frequency ranges and typical applications.

**Q10.** A network administrator needs to connect 6 offices in a star topology through a central hub. Draw the topology, calculate the number of links required, and justify why star topology was chosen over mesh for this scenario.

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--- End of Assignment ---